

Started on Sunday, 6 April 2025, 2:31 PM

State Finished

Completed on Sunday, 6 April 2025, 3:34 PM

Time taken 1 hour 3 mins

Question 1

Complete

Marked out of 1.50

If a constraint graph has no loops, then the CSP can be solved in _____ time?

- ☒ A. $O(d^n)$
- ☐ B. $O(nd^2)$
- ☐ C. $O(n^2d)$
- ☐ D. $O(n^d)$

Question 2

Complete

Marked out of 1.50

Which of the following methods are used to split the data?

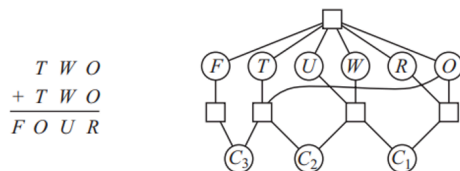
- ☒ A. Random Split
- ☒ B. K-fold split
- ☒ C. Stratified split
- ☒ D. Leave-One-Out Cross-Validation

Question 3

Complete

Marked out of 1.50

A cryptarithmic problem. Each letter stands for a distinct digit; the aim is to find a substitution of digits for letters such that the resulting sum is arithmetically correct, with the added restriction that no leading zeros are allowed.



Which of the following constraints are correct for modeling this problem?

- ☒ A. $O + O = R + 10.C1$
- ☒ B. $C1 + W + W = U + 10.C2$
- ☒ C. $T > 0, F > 0$
- ☐ D. $F = C3 = 1$

Question 4

Complete

Marked out of 2.00

In simulated annealing, if the temperature T is set to zero, the algorithm behaves like:

- ☐ a. Greedy Best-First Search
- ☐ b. Uniform Cost Search
- ☒ c. Hill-Climbing Search
- ☐ d. Random Walk

Question 5

Complete

Marked out of 2.00

Consider an objective function over an 8×8 grid where each cell represents a state. If 15% of states are local maxima and you perform steepest-ascent hill climbing from a randomly chosen initial state, what is the probability of getting stuck in a local maximum (that is not the global maximum)? Assume only one state is the global maximum.

- ☐ a. 0.156
- ☐ b. 1
- ☐ c. 0.85
- ☒ d. 0.140

Question 6

Complete

Marked out of 2.00

Which of the following best distinguishes simulated annealing from hill-climbing?

- ☐ a. Simulated annealing uses a stack
- ☒ b. Simulated annealing accepts worse moves with a probability that decreases over time
- ☐ c. Simulated annealing avoids plateaus using random restarts
- ☐ d. Simulated annealing only works on discrete spaces

Question 7

Complete

Marked out of 1.50

Which of the following is not a constraint satisfaction problem?

1. Sudoku
2. 8 Queens puzzle
3. Maximum Cut problem

- ☐ A. 1 and 2
- ☒ B. 3 only
- ☐ C. All of the above
- ☐ D. None of the above

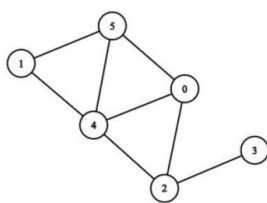
Question 8

Complete

Marked out of 1.50

In the following constraint graph, each variable can take value from set {a,b,c}, given the constraint such that each node connected through an edge should have a different value. Use the minimum value remaining heuristic to find the order of assignments of the nodes.

Note: For node tie breaking follow sorted numeric ordering and for value tie breaking follow lexicographical order. Your answer should be in a sequence of <node, value> pair such as 1a2c.



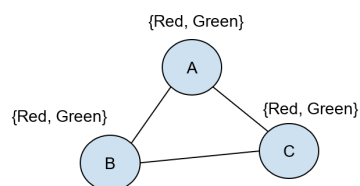
4a0b5c1b2c3a

Question 9

Complete

Marked out of 1.50

Is the basic Arc Consistency algorithm able to identify the failure for the below case ? **(yes/no)**



- ☐ True
- ☒ False

Question 10

Complete

Marked out of 1.50

Perform forward checking on the following graph and identify which region left with no legal values after $V = \text{blue}$?



	WA	NT	Q	NSW	V	SA	T
Initial domains	R G B	R G B	R G B	R G B	R G B	R G B	R G B
After $WA = \text{red}$	(R)	G B	R G B	R G B	R G B	G B	R G B
After $Q = \text{green}$	(R)		B	(G)	R B		B R G B

- ☒ A. SA
- ☐ B. NSW
- ☐ C. NT
- ☐ D. T

Question 11

Complete

Marked out of 1.50

Suppose we have a CSP with n nodes and make it strongly n -consistent (i.e., strongly k -consistent for $k = n$) and each variable can take d values, we are guaranteed to find a solution in time _____

- ☒ A. $O(d^n)$
- ☐ B. $O(nd^2)$
- ☐ C. $O(n^2d)$
- ☐ D. $O(n^d)$

Question 12

Complete

Marked out of 1.50

Identify which type of constraint the following represents?

- Indians won't tolerate the color black.
- Asserting that the value of Y is between X and Z .
- All of the variables involved in the constraint must have different values.
- Apple is better than Grapes.

- ☐ A. 1. binary 2. ternary 3. higher order 4. binary
- ☐ B. 1. binary 2. higher order 3. global 4. soft
- ☒ C. 1. unary 2. ternary 3. global 4. soft
- ☐ D. 1. soft 2. ternary 3. higher order 4. binary

Question 13

Complete

Marked out of 2.00

In gradient descent on a continuous function $f(x) = x^2 + 4x + 4$, starting at $x_0 = 5$ with a learning rate $\alpha = 0.1$, what will be the value of x after one update step? [Cheat: The gradient of $f(x)$ is $f'(x) = 2x + 4$ and the update rule is $x' = x - \alpha \times f'(x)$]

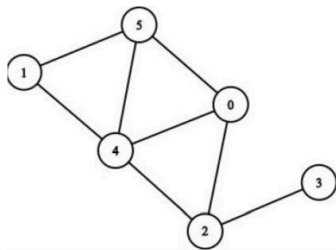
- ☐ a. 4.6
- ☐ b. 1.4
- ☒ c. 3.6
- ☐ d. 5.0

Question 14

Complete

Marked out of 1.50

To solve the constraint satisfaction problem, which variable would you choose first from the following constraint graph that will simplify the search procedure? **(answer should be a number)**



4

Question 15

Complete

Marked out of 1.50

What is a "plateau" in the context of hill-climbing?

- ☐ a. A state with the highest possible value
- ☒ b. A flat region with identical heuristic values for neighbouring states
- ☐ c. A sequence of states leading to the global maximum
- ☐ d. A region with no valid neighbouring states

Question 16

Complete

Marked out of 2.00

What is the most significant drawback of simple hill-climbing search?

- ☐ a. High space complexity
- ☐ b. Inability to find a goal
- ☒ c. Getting stuck in local maxima
- ☐ d. It cannot operate in continuous state spaces

Question 17

Complete

Marked out of 2.00

In a simulated annealing algorithm, a move that modifies the objective function value by 8 units is being considered at a temperature of $T = 5$. Using the acceptance probability formula $P(\text{accept}) = e^{(-\Delta E/T)}$, what is the probability (rounded to two decimal places) that this worse move will be accepted?

- ☒ a. 0.20
- ☐ b. 0.80
- ☐ c. 0.27
- ☐ d. 0.13

Question 18

Complete

Marked out of 1.50

Which of the following is true for the CSPs?

1. Tree structured CSPs cannot be solved in polynomial time.
2. Arc consistency can identify failures in every case.
3. Computation time depends on number constraints and variables?
4. Local search can attempt to reduce unsatisfied constraints by the min-conflicts heuristic.

- ☐ A. Only 2 and 4
- ☐ B. Option 1, 2, 4
- ☒ C. Only 3 and 4
- ☐ D. Only 4